Inhibitory population geometry in perceptual multistability

Dissociating the precision of sensory evidence from the stability of the prior, and their offline affective reweighting

Research proposal · Interdisciplinary Research Master · November 2026

Abstract.

When sensory input is ambiguous, perception does not settle: it alternates spontaneously between mutually exclusive interpretations. This perceptual multistability is conserved from *Drosophila* to humans, is altered in a transdiagnostic fashion across psychiatric and neurodevelopmental conditions, and is widely attributed to a disturbed "excitation–inhibition balance." That attribution is almost certainly too coarse. Inhibition is not one quantity but a set of functionally distinct interneuron classes whose collective activity occupies structured, low-dimensional subspaces, and recent work argues that only a population-level description can capture how those classes cooperate. This proposal asks whether the two cardinal behavioural signatures of multistability — how *exclusive* a percept is, and how *stable* it is — are controlled by dissociable interneuron subspaces, and whether the affective value of a percept is written into that geometry offline, during sleep. In head-fixed mice viewing ambiguous moving plaids, I will (1) map the population geometry of simultaneously recorded PV-, SOM- and VIP-expressing interneurons onto latent parameters of Bayesian and reinforcement-learning models fitted to switching behaviour; (2) test causal necessity by driving interneuron activity along versus orthogonal to the identified communication subspace, a contrast that yields an unusually sharp falsification; and (3) pair one percept with an aversive outcome and ask whether its cortical assembly is preferentially reactivated during NREM sleep, whether that reactivation predicts the next day's perceptual prior, and whether this mechanism is specifically altered in MECP2-duplication mice. The project replaces a single scalar — E/I ratio — with a cell-type-resolved, geometric account of perceptual inference, and in doing so converts a cheap, language-free behavioural assay into a candidate readout of circuit-specific pathology.

1. Problem definition and hypotheses

The problem. Ambiguous stimuli produce alternating percepts, and the statistics of that alternation — dominance durations, switch rate, the incidence of "mixed" percepts in which no single interpretation wins — differ reliably between clinical and non-clinical populations [22]. Two observations frame the field's current explanation. First, visual-cortical GABA concentration correlates with the suppression of mixed percepts in non-clinical participants [19], and pharmacologically increasing GABAergic signalling reduces them [17]. Second, altered excitation–inhibition balance is a leading pathophysiological account of psychosis and of autism [11]. Together these have made "E/I imbalance" the default mechanistic story for atypical multistability.

The story cannot be right as stated. It treats inhibition as a scalar gain, whereas cortical inhibition is delivered by interneuron classes with distinct targets, kinetics and modulatory inputs, and the behavioural phenotypes are *dissociable*: autistic adults show both slower switching and more mixed percepts, whereas several psychiatric conditions show increased stability without an excess of mixed percepts, and anxiety disorders diverge again [22]. A single dial cannot produce dissociable effects. Worse, bulk measures such as magnetic-resonance spectroscopy of GABA and systemic pharmacology are precisely the measurements that cannot distinguish among interneuron classes. The field therefore has a mechanistic hypothesis pitched at a level of description that its own measurements cannot resolve (Fig. 1a).

The opportunity. Letzkus and colleagues [14] argue that interneuron research has exhausted what the single-neuron and microcircuit approaches can deliver and must move to the population level, where the relevant object is not a cell's tuning but the *geometry* of collective activity: the dimensionality and curvature of the manifold on which population activity lives, the decoding axes along which task variables are read out, and the *communication subspaces* through which one

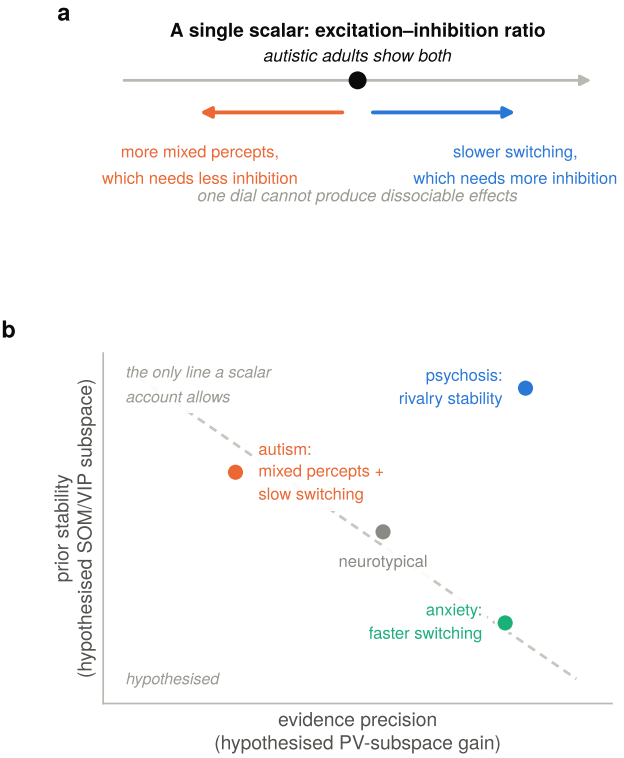


Figure 1. From a scalar to a plane. (**a**) A scalar excitation–inhibition account offers one axis, but autistic adults show both an excess of mixed percepts and slower switching, which that axis requires to lie in opposite directions. (**b**) The two-parameter account proposed here. Phenotype positions are hypothesised, not measured; the dashed line marks the one-dimensional family of states a scalar account can represent.

population transmits a selected subset of its activity to another while keeping the orthogonal remainder private [12]. Critically, they note that recent work already hints at a division of labour: in hippocampus, PV-cell activity correlates with place fields, SOM with context, and VIP with reward [15, 8]. Mapped onto perceptual inference, those three correlates read almost like a translation table — sensory content, prior context, and value. That mapping is the hypothesis this proposal tests.

Central hypothesis. Perceptual multistability is governed not by a scalar E/I ratio but by the geometry of interneuron population activity: functionally distinct interneuron classes occupy separable subspaces that carry dissociable computational variables of perceptual inference, and the affective value of a percept is consolidated into that geometry offline, during NREM reactivation.

Five testable predictions follow.

H1 — Exclusivity. PV population gain along its dominant percept-encoding axis sets the *exclusivity* of the percept, i.e. the precision of the sensory likelihood. Predicted: PV population magnitude correlates negatively with mixed-percept probability; PV populations carry the highest percept-identity decoding weights of the three classes.

H2 — Stability. SOM and VIP populations set the *stability* of the current percept — prior precision and the drive toward an "internal action" that terminates it [20, 21]. Predicted: perceptual switches are preceded, by hundreds of milliseconds, by a rotation of SOM/VIP population activity into the communication subspace onto pyramidal and PV populations, such that impending switches are decodable before the behavioural report (Fig. 2a).

H3 — Null-space specificity. Only activity *aligned* with the identified communication subspace is behaviourally consequential. Predicted: equal-magnitude, equal-energy perturbation along an orthogonal direction is behaviourally silent. This is the proposal's sharpest falsification: if orthogonal and aligned perturbations are equally effective, the geometric framework is descriptive rather than causal, and H1–H2 lose their mechanistic content.

H4 — Offline affective reweighting. After one percept is paired with an aversive outcome, that percept's cortical assembly is preferentially and more faithfully reactivated during subsequent NREM sleep, preferentially within cortico-hippocampal coordinated sharp-wave-ripple windows, and the magnitude of that reactivation predicts the next day's shift in the perceptual prior [18].

H5 — Clinical specificity. In MECP2-duplication mice, which show atypical visual rivalry [1], the phenotype maps onto a *specific* geometric parameter — predicted: reduced PV-subspace gain with preserved or increased prior-subspace stability — rather than a uniform reduction of inhibitory drive. H5 is the point at which "cell-type-resolved geometry" and "global E/I imbalance" make different predictions from the same behavioural data.

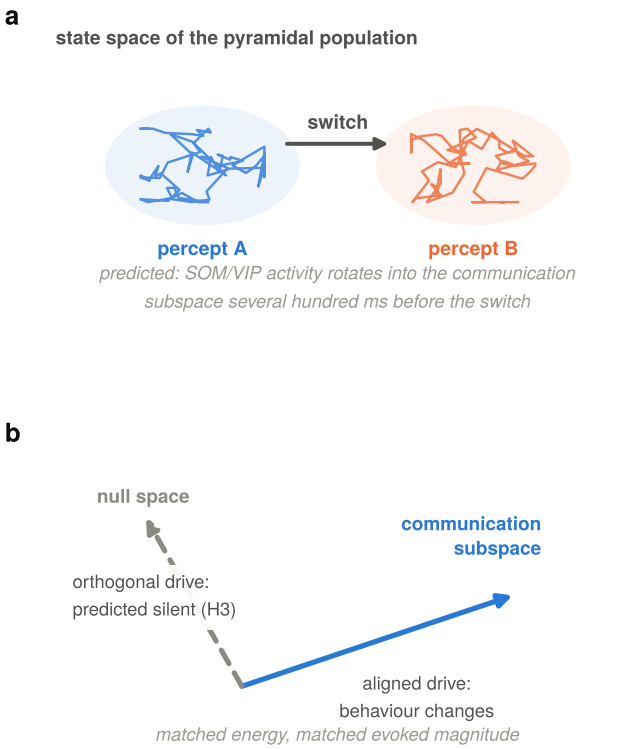


Figure 2. Hypothesised geometry and the causal test. (**a**) Population activity dwells in one of two regions of state space and transitions spontaneously; H2 predicts that SOM/VIP activity rotates into the communication subspace before the switch. (**b**) The Aim 2 contrast. Drive aligned with the communication subspace should change behaviour; equal-energy drive along the null space should not (H3). Both panels are schematic.

2. Specific aims

Aim 1 (map). Determine whether PV, SOM and VIP populations in mouse V1 and posterior parietal cortex carry dissociable computational variables during spontaneous perceptual alternation, by relating population geometry (decoding axes, manifold curvature, communication subspaces, single-cell decoding weights) to latent parameters of Bayesian and reinforcement-learning models fitted to behaviour. *Tests H1, H2.*

Aim 2 (perturb). Establish causal necessity and subspace specificity by driving identified interneuron ensembles along versus orthogonal to the communication subspace identified in Aim 1, with matched photostimulation energy and matched evoked population magnitude. *Tests H2, H3.*

Aim 3 (consolidate, and break). Pair one percept with an aversive outcome and determine whether offline reactivation of its cortical assembly during NREM sleep predicts the subsequent shift in the perceptual prior, and whether this mechanism is selectively altered in MECP2-duplication mice. *Tests H4, H5.*

The aims are sequenced so that each supplies the other's parameters: Aim 1 defines the subspaces Aim 2 perturbs, and Aims 1–2 define the assemblies whose reactivation Aim 3 tracks. Aim 1 is therefore the only

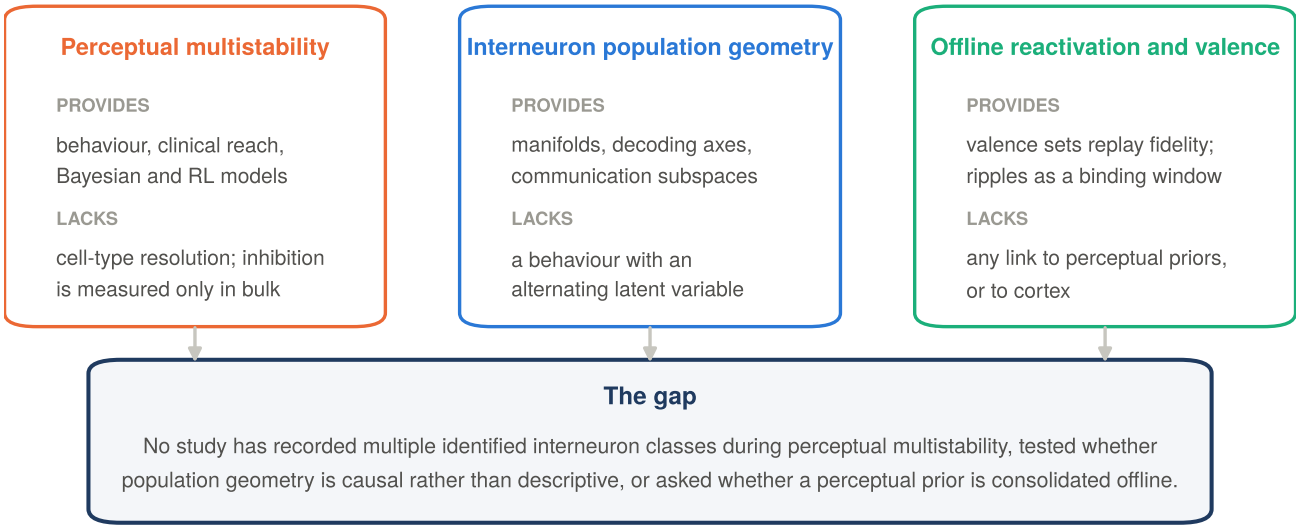


Figure 3. Three literatures and the gap between them. Each of the three source studies supplies what the other two lack. The proposal is positioned at their intersection.

aim on the critical path, and it is the least technically risky.

3. Background and previous work

Each of the three source literatures stops where the next begins (Fig. 3).

Multistability as a model system. Perceptual multistability has been documented in *Drosophila*, zebrafish, reptiles, pigeons, mice, non-human primates and humans [3], with dominance durations following gamma-like distributions and comparable psychophysical dependencies across paradigms [22]. Its cross-species reach is what makes it unusual: the same behavioural measure can be obtained from a patient in a clinic and from a mouse under a two-photon microscope. Mouse work is now established — ambiguous moving plaids elicit alternation in mice, and switch rates are altered in the MECP2-duplication model of autism [1] — and population dynamics during binocular conflict have been described in mouse visual cortex [24]. Human work has generated a mature computational scaffolding: Bayesian accounts treat switching as sampling from, or exploration of, a posterior with two modes, with clinical differences arising from mis-weighted priors or likelihoods [23, 4] or from the overcounting of evidence in circular inference [5, 13]; and a decision-theoretic account treats each switch as an internal action chosen against an opportunity cost, making multistability a form of internal foraging [20, 21]. Safavi and colleagues argue that the decisive next step is to combine these frameworks with animal models, since only there can behaviour be connected to the underlying circuitry [22]. They are explicit that the required experimental approaches do not yet exist, listing as an outstanding question what methods are needed "to elucidate the fine-grained functional roles of the large, distributed neural networks" involved.

The population approach to interneurons. The missing method is plausibly the one proposed by the IN-CODE consortium [14]. Interneurons constitute only 15–20% of cortical neurons, so population-level analysis of identified interneuron classes was long infeasible; enhancer-AAV toolboxes [7], post-hoc neurochemical identification of imaged populations [2], and large-scale imaging of molecularly identified interneurons [8] have removed that barrier. Letzkus et al. set out the resulting analytic programme: decode task variables from each interneuron type's population activity; characterise manifold geometry and its changes with state and experience; identify communication subspaces between interneuron types and pyramidal cells; and correlate per-neuron decoding weights with molecular and physiological identity, thereby bridging population codes and single-cell signatures. They also flag two difficulties this proposal must confront directly: interneuron cell counts and firing rates differ systematically across classes, so it is genuinely unresolved whether decoding should be compared at matched cell number, normalised to tissue prevalence, or normalised to emitted spikes; and within-type heterogeneity may be as functionally important as between-type differences [26].

Offline reactivation and valence. The third component concerns how a perceptual prior is *changed*. Training biases dominance toward trained features [22], which implies a learning process, and learning of this kind is consolidated offline. Morici and colleagues [18] recorded simultaneously across the dorsoventral hippocampal axis in rats performing the same spatial task under aversive or rewarding valence. Joint dorsoventral assemblies, detected by independent component analysis [16], disambiguated the two valence conditions better than purely dorsal or ventral assemblies; both sets reactivated during post-task NREM sleep with activation rates decaying over the first 40 minutes; and crucially, reactivation of *aversive* assemblies mirrored the orig-

inal wake patterns more faithfully than reward assemblies, an effect driven by increased recruitment of ventral shock-responsive neurons into coordinated dorsal–ventral ripples and by enriched dorsal place-cell replay in those same windows. Coordinated ripples thus constitute a privileged window in which spatial and affective representations are bound with heightened fidelity. Hippocampo-cortical circuits are known to route and replay selectively [10], and ventral hippocampal interneurons carry anxiety-related activity [6], so the same logic can be asked of a cortical perceptual assembly: does aversive tagging of one interpretation of an ambiguous stimulus produce privileged, high-fidelity offline reactivation of that interpretation's assembly, and is that what sets the next day's prior?

The gap. Each literature stops exactly where the next begins. Multistability research has the behaviour, the clinical reach and the computational models, but measures inhibition only in bulk. The interneuron-population literature has the cell-type resolution and the geometric analytics, but has not been applied to a behaviour in which a latent variable demonstrably alternates under constant input — arguably the cleanest available assay of internally generated inference. The reactivation literature has established that valence biases the fidelity of offline replay, but in hippocampus and for spatial memory, not for perceptual priors. No study has recorded multiple identified interneuron classes during perceptual multistability; none has tested whether population geometry is causally rather than descriptively related to perceptual switching; and none has asked whether a perceptual prior is consolidated offline.

4. Significance

Within the field. The proposal would replace the field's current mechanistic hypothesis with one pitched at a level of description that can actually be measured. If H1 and H2 hold, "E/I imbalance" resolves into at least two independent parameters with distinct cellular substrates and distinct behavioural signatures, which would explain why multistability phenotypes dissociate across conditions instead of lining up on a single axis. H3 addresses a question larger than multistability: population geometry is now the standard descriptive language of systems neuroscience, yet demonstrations that activity *within* an identified communication subspace matters while equal activity orthogonal to it does not remain rare. A clean null-space test in a behaviour with a trial-by-trial latent variable would be of general methodological interest. Aim 1 also delivers a pre-registered answer to the normalisation problem Letzkus et al. leave open, which is a nuisance affecting every comparison of decoding across populations of unequal size.

Outside the field. Three transfers are plausible. (i) *Clinical*. Multistability requires no language, no instruction beyond "report what you see," and no expensive hardware; it is already proposed as a transdiagnostic assay [22]. If switch rate and mixed-percept rate index dissociable interneuron parameters, a five-minute behavioural profile becomes a candidate cell-type-specific readout, which is the kind of stratification that trials of E/I-modulating compounds currently lack. (ii) *Machine learning and interpretability*. The dominant method for understanding learned representations is to identify latent directions and assume they are functionally meaningful; H3 is the biological version of the causal test that this assumption usually escapes, and the aligned-versus-orthogonal design transfers directly to activation steering in artificial networks. (iii) *Measurement theory in psychology*. If a behavioural statistic can be shown to index a specific circuit parameter rather than a diffuse construct, that is a worked example of the construct validity that computational phenotyping promises and rarely demonstrates.

5. Research design and methods

All work uses adult mice of both sexes, head-fixed on a cylindrical treadmill, water-restricted, and trained on an ambiguous-motion task adapted from Bogatova et al. [1]. Animals first learn to report the direction of an unambiguous moving plaid or structure-from-motion sphere by licking one of two spouts; ambiguous stimuli are then introduced in test blocks. Stimulus ambiguity is varied parametrically over five levels of depth-cue strength, which manipulates likelihood precision independently of the prior. Each session includes **no-report blocks** in which the ambiguous stimulus is viewed without a required response, with optokinetic nystagmus and pupil dynamics as proxies for the current percept, so that neural correlates of the percept can be separated from correlates of the report [25]. Behaviour is characterised by dominance- duration distributions (gamma fits), switch rate, and mixed-percept proportion, and is fitted with both a hierarchical Bayesian/circular-inference model [13] and a decision-theoretic internal-foraging model [21]; trial-by-trial latent estimates from these fits are the regressors against which geometric measures are tested. Figure 4 states the pattern of results each aim predicts.

Aim 1 — Mapping interneuron population geometry onto inference variables

Design. Ten C57BL/6J mice, approximately eight imaging sessions each, V1 and posterior parietal cortex. Spectrally distinct calcium indicators are expressed in PV-, SOM- and VIP-expressing interneurons using enhancer-AAVs [7] alongside a pan-neuronal indicator for pyramidal populations, with three-colour two-photon imaging and post-hoc immunohistochemical confirmation of class identity in a subset of fields [2]. Target yields per session are at least 300 pyramidal, 40 PV, 40 SOM and 30 VIP neurons; where three-colour imaging proves unreliable, two classes are imaged per session and the third interleaved across sessions.

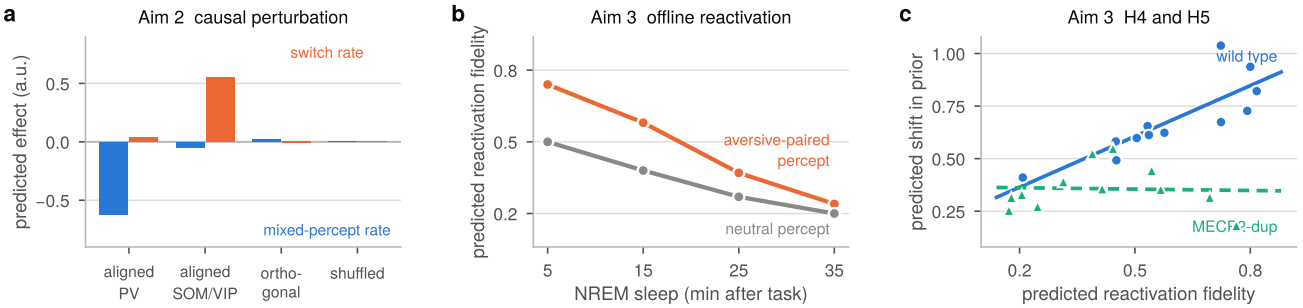


Figure 4. Predicted outcomes. No data shown; every panel is a prediction. (**a**) Aim 2: aligned PV drive should suppress mixed percepts, aligned SOM/VIP drive should shift switch rate, and orthogonal drive should resemble the shuffled control. (**b**) Aim 3: the aversively paired percept's assembly should reactivate with higher fidelity, with both conditions decaying over the first 40 minutes of NREM sleep. (**c**) Aim 3: reactivation fidelity should predict the next day's shift in the prior in wild-type animals (H4), and that relationship should be degraded in MECP2-duplication mice (H5).

Analysis. (i) Cross-validated linear decoding of percept identity and of impending switches from each population separately. Because decoding accuracy depends jointly on neuron number and firing rate [14], every comparison is reported in three pre-registered forms — matched cell number, matched event rate, and prevalence-normalised — with the matched-cell-number version as the primary analysis and divergence among them treated as a finding rather than a nuisance. (ii) Manifold geometry: participation ratio, curvature, separation of the two percept basins, and drift trajectories in the seconds preceding spontaneous switches. (iii) Communication subspaces between each interneuron population and pyramidal populations by reduced-rank regression, with recurrent network fits where populations are mutually coupled. (iv) Per-neuron decoding weights regressed against physiological properties and post-hoc molecular identity, to test whether functional signatures align with the standard taxonomy or cut across it.

Expected outcomes and alternatives. H1 and H2 predict a double dissociation: PV decoding weights dominate percept identity and PV population magnitude tracks mixed percepts, whereas SOM/VIP subspace alignment tracks switch timing. If instead all three classes carry equivalent information with equivalent weights, the division-of-labour hypothesis fails at this granularity, and the analysis pivots to within-type heterogeneity — ordering cells along a transcriptomic or physiological axis [2, 26] and asking whether that axis, rather than class membership, predicts decoding weight. This alternative is not a salvage operation: it is the second of the two gaps Letzkus et al. identify, and a negative result for between-type specialisation combined with a positive result for within-type ordering would be a substantive finding.

Aim 2 — Causal test of subspace specificity

Design. Eight mice, within-animal design, approximately fifteen sessions each. A soma-targeted excitatory opsin is delivered by the same class-specific enhancer-AAVs and driven by spatial-light-modulator holographic stimulation of 20–50 identified neurons, interleaved with imaging of all four populations. On each perturbation trial the stimulated ensemble is chosen so that its population weight vector is (a) aligned with the communication subspace identified in Aim 1, (b) orthogonal to it, or (c) a shuffled pattern matched for mean drive. Total photostimulation energy is matched across conditions, and — more importantly — conditions are additionally matched on the *measured* evoked population magnitude, so that the aligned/orthogonal contrast cannot be explained by a difference in how much activity was injected.

Analysis and predictions. Shifts in switch rate, dominance-duration distribution and mixed-percept proportion, and shifts in the fitted model parameters. Aligned PV drive should increase exclusivity and reduce mixed percepts; aligned SOM/VIP drive should shift switch rate with a sign depending on the direction of the perturbation along the switch-predictive axis; orthogonal drive should be behaviourally silent (H3). Off-target network effects are measured directly, since the non-targeted populations are imaged concurrently.

Pitfalls. Opsin–indicator spectral crosstalk is addressed by spectral separation and sham-stimulation controls at matched laser power. The principal risk is that any strong interneuron drive saturates behaviour, collapsing the aligned/orthogonal contrast; this is mitigated by titrating to the lowest effective photostimulation power in pilot sessions and by including a graded dose series rather than a single intensity.

Aim 3 — Offline affective reweighting, and its disruption

Design. Twelve wild-type and twelve MECP2-duplication mice. This aim switches from two-photon imaging to chronic Neuropixels recordings in V1, posterior parietal cortex and dorsal CA1 — the temporal resolution of sharp-wave ripples and sleep dynamics requires electrophysiology — with optogenetic tagging of PV, SOM and VIP units for cell-type identification. Recording spans a pre-task sleep session, the task, and a post-task sleep session. One percept (counterbalanced across animals) is paired with a mild aversive air puff; the alternative percept remains neutral.

Analysis. Following Morici et al. [18]: ICA-based

assembly detection during wake [16]; assembly activation peak rates in pre- versus post-task NREM; reactivation *fidelity*, defined as the similarity between the reactivated pattern and the original wake pattern; assembly gain during cortico-hippocampal coordinated ripple windows versus isolated ripples; and the relationship between wake recruitment strength and reactivation fidelity. The behavioural readout is the next-day shift in initial-percept probability and dominance duration, i.e. the measured change in the prior, and the key test of H4 is whether reactivation fidelity predicts that shift across animals and sessions. H5 is tested by asking which geometric parameter differs by genotype, and whether that parameter is the one that Aim 1 associated with the behavioural phenotype the genotype displays.

Stretch component. If time permits, closed-loop disruption of coordinated ripple windows during post-task sleep [9] would test whether offline reactivation is *necessary* for the prior shift. This is flagged as optional and is not load-bearing for any hypothesis.

Power and statistical approach

Group comparisons in Aim 3 are powered for a between-genotype effect of $d = 1.2$, the approximate magnitude of the reported switch-rate difference in MECP2-duplication mice [1], giving approximately 80% power at $n = 12$ per group and $\alpha = 0.05$. Within-animal perturbation contrasts in Aim 2 are powered for $d = 0.8$ at roughly fifteen sessions per animal. These are planning estimates to be refined against pilot variance before the confirmatory cohorts are run, which is the function of the go/no-go gates in §9. Throughout, animal is treated as a random effect in mixed models rather than pooling across sessions, hypotheses H1–H5 are pre-registered with their primary analyses specified, and both the raw and the normalisation-variant analyses are reported regardless of outcome.

6. Preliminary evidence and feasibility

No pilot data exist for this project; the evidence that it is feasible and that its predictions are non-trivial comes from quantitative reading of the three source studies, and the proposal's projections are derived from their reported effects.

Feasibility of the behaviour. Bogatova et al. established alternation to ambiguous plaids in mice and detected a genotype difference in switch rate, which fixes both the paradigm and the effect size used in the power calculation [1]. The risk is not whether mice alternate but whether they can report alternation reliably enough for trial-by-trial model fitting; the no-report blocks exist so that Aim 1's core analyses survive a failure of the report contingency.

Feasibility of the recordings. The three-colour requirement is the most demanding element. Bugeon et al. [2] and Geiller et al. [8] demonstrate population-scale recording of molecularly identified interneurons, and enhancer-AAV toolboxes now provide class-specific access with soma-targeted payloads [7], but yields of 30–40 cells per class per session are at the upper end of what is routine, which is why the two-class interleaved fallback is specified in advance rather than improvised.

Quantitative anchoring of Aim 3. Morici et al. report that reactivation declines markedly over the first 20–40 minutes of NREM sleep and that aversive, but not reward, assemblies show a correlation between wake recruitment strength and reactivation fidelity [18]. Two design consequences follow: post-task sleep analysis windows are fixed a priori at 0–20 and 20–40 minutes of NREM rather than chosen post hoc, and the aversive condition — not a reward condition — is the one in which H4 can realistically be detected, which is why valence is manipulated as aversive-versus-neutral rather than aversive-versus-rewarding. Their observation that reward effects may have been masked by task familiarity also implies that the perceptual conditioning in Aim 3 should be introduced as a novel contingency rather than after extended overtraining.

Interpretation of the discrepant evidence. The source literatures are not fully consistent, and the proposal takes a position. Increased perceptual stability in binocular rivalry is reported across several psychiatric conditions and is most readily explained by *overweighted* priors, whereas reduced reliance on hard-wired priors in schizophrenia and autism points to *attenuated* priors [22]. A single scalar E/I account must choose; a two-parameter geometric account need not, because reduced PV-subspace gain (lower evidence precision) and increased prior-subspace stability can co-occur, and their combination predicts exactly the otherwise paradoxical pairing of more mixed percepts with slower switching seen in autistic adults. H5 is the direct test of that reconciliation.

7. Critical considerations and limitations

Assumptions that could fail. Three load-bearing assumptions deserve to be named. First, that mouse perceptual alternation and human multistability share mechanisms; the cross-species conservation of the phenomenon [3] supports this but does not establish it, and a negative result in mice would be uninformative about human psychiatry. Second, that the standard PV/SOM/VIP taxonomy is the right granularity — the issue Letzkus et al. raise explicitly, and the reason within-type heterogeneity is built in as a planned alternative rather than a post-hoc excuse. Third, that holographic stimulation of 20–50 cells is sufficient to move a population along a chosen subspace direction without recruiting the network so broadly that the aligned/orthogonal distinction dissolves; this is an empirical question that the dose series in Aim 2 is designed to answer, and a null result there would be uninterpretable rather than negative, which is an honest weakness.

What would falsify the framework. H3 is the decisive test. If equal-energy, equal-magnitude pertur-

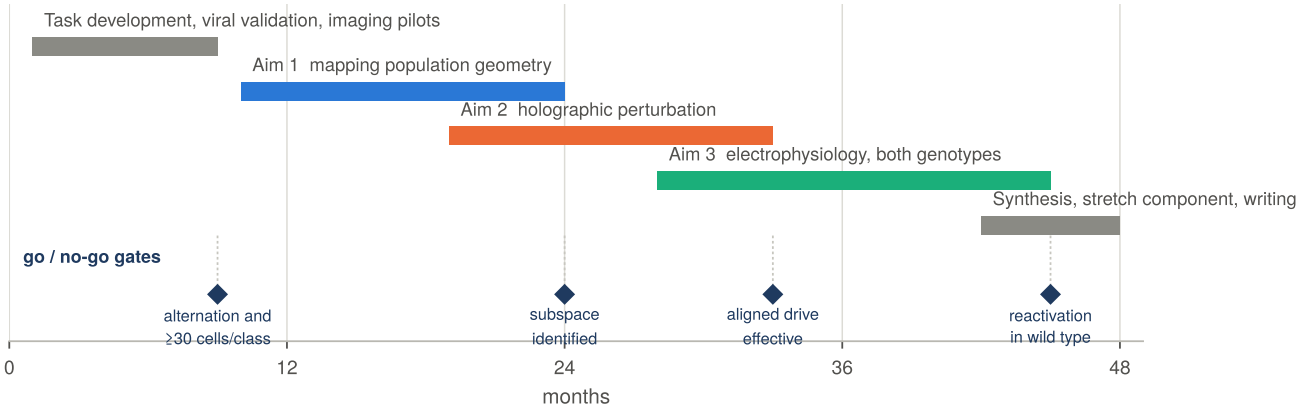


Figure 5. Timeline and go/no-go gates. Each gate converts a possible failure into a narrower but still publishable result: the confirmatory cohort is not run until pilots show reliable alternation and adequate per-class yield; Aim 2 is not scaled until aligned drive is shown to be behaviourally effective; and the genotype comparison is not attempted until assembly reactivation is detectable in wild-type animals.

bations aligned with and orthogonal to the communication subspace have indistinguishable behavioural effects, then population geometry is a convenient description of these data rather than the level at which the circuit operates, and the proposal's central claim fails even if H1 and H2 survive as correlations. Stating that in advance is deliberate: the framework's appeal is that it is specific enough to be wrong.

Scope limits. The project addresses visual multistability in rodents and a single genetic model; it does not address auditory or multistable-figure paradigms, does not speak to the frontoparietal contributions that dominate the human literature [22], and cannot test the dopaminergic and neuromodulatory hypotheses that Safavi et al. list as outstanding. Those are deliberate exclusions in favour of a tractable first mechanistic test, and the proposed open data release is intended to let others address them.

8. Broader impacts

Clinical and translational. The immediate translational target is stratification rather than treatment. If switch rate and mixed-percept rate index dissociable circuit parameters, a brief language-free behavioural assay could identify which patients have a precision deficit versus a prior-stability deficit, a distinction that currently has no cheap behavioural proxy and that matters for any intervention acting on specific interneuron populations. The assay's independence from language and instruction also makes it usable in populations that psychometric instruments serve poorly, including minimally verbal autistic people and young children.

Methodological and cross-disciplinary. The aligned-versus-orthogonal perturbation design is a general template for testing whether an identified latent direction is causally load-bearing, and it transfers to activation steering and interpretability work in artificial networks, where the analogous causal test is rarely run. The pre-registered resolution of the decoding-normalisation problem is a small but field-wide contribution, since the same confound affects every comparison of information content across populations of unequal size.

Open science and training. All analysis code will be released, with raw electrophysiology and imaging deposited in public repositories, and the behavioural task and stimulus code released so that the paradigm can be adopted without reimplementation. The project is itself interdisciplinary training in the strict sense: it requires competence in rodent electrophysiology and imaging, in Bayesian and reinforcement-learning modelling, and in the clinical literature on perceptual atypicality, and it is designed so that none of those three can be treated as a service component.

Ethical. The use of an aversive stimulus is the proposal's main ethical cost and is deliberately minimised: a mild air puff rather than a shock, the smallest number of pairings that establishes a contingency, and valence manipulated as aversive-versus-neutral rather than requiring a second aversive condition. Work will be conducted under EU Directive 2010/63/EU with institutional animal welfare body and national competent authority approval, with explicit humane endpoints, and with reduction achieved by the within-animal designs in Aims 1 and 2, which allow the group sizes above to support the proposed contrasts.

9. Timeline, go/no-go gates and dissemination

Aim 2 begins before Aim 1 concludes, using interim subspace estimates, so that the dependency does not serialise the whole project; the gate structure means a failure at any stage converts into a narrower but still publishable result rather than a dead end.

Expected outputs. Three first-author papers are anticipated — a methods-and-mapping paper from Aim 1, a causal paper from Aim 2, and a consolidation-and-model paper from Aim 3 — each preceded by a preprint. Aim 2 is well suited to submission as a registered report, given that its predictions are directional and specified in advance. Dissemination will include presentations at FENS, the Bernstein Conference, Cognitive Computational Neuroscience and the Society for Neuroscience, with the clinical framing presented to a computational-psychiatry rather than a systems-

neuroscience audience at least once, since the transdiagnostic claim needs scrutiny from people who see patients. The proposal will be defended orally with the H3 falsification test as the organising argument, on the view that a proposal is best judged by what would refute it.

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Appendix A — Mapping to the evaluation rubric

Criterion	Where addressed
1. Problem definition / hypothesis	§1: the E/I-scalar problem, central hypothesis, H1–H5 (Fig. 1)
2. Specific aims	§2: three aims with their dependency structure
3. Background and previous work	§3: the three source literatures and the gap between them (Fig. 3)
4a. Significance within the field	§4, first paragraph
4b. Significance outside the field	§4, second paragraph: clinical, ML interpretability, measurement theory
5. Research design and methods	§5: per-aim design, analysis, predictions and pitfalls (Figs. 2, 4); power and statistics
6. Results / preliminary studies	§6: quantitative interpretation of the source data and the design consequences drawn from it
7a. Written communication	Throughout; §7 states what would falsify the framework
7b. Oral communication	§9: defence organised around the H3 falsification test
8. Critical thinking	§6 (reconciling discrepant findings), §7 (named assumptions, scope limits, an honest uninterpretable-null case)
9. Broader impact	§8: clinical, methodological, open science, training, ethical
10. Publications and presentations	§9: three planned papers, preprints, registered report, conference plan (Fig. 5)